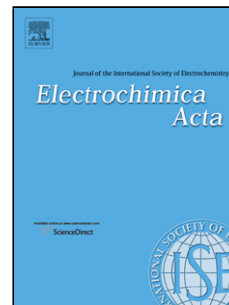


## Accepted Manuscript

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PII: S0013-4686(17)31071-X  
DOI: <http://dx.doi.org/doi:10.1016/j.electacta.2017.05.075>  
Reference: EA 29510

To appear in: *Electrochimica Acta*

Received date: 29-3-2017  
Revised date: 12-5-2017  
Accepted date: 12-5-2017

Please cite this article as: W. Hu, C. Zhang, H. Jiang, M. Zheng, Q.-H. Wu, Q. Dong, Improving the electrochemistry performance of layer  $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$  material at 4.5 V cutoff potential using lithium metaborate, *Electrochimica Acta* (2017), <http://dx.doi.org/10.1016/j.electacta.2017.05.075>

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## Improving the electrochemistry performance of layer $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$ material at 4.5 V cutoff potential using lithium metaborate

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### Abstract:

$\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$  (NMC) cathode material is surface treated by Li active boron oxide (LBO) via a two-step process. The surface morphology, crystal structure, surface composition, and electrochemical properties were characterized based on SEM, XRD, XPS, charge–discharge tests, cyclic voltammetry (CV), and electrochemical impedance spectroscopy (EIS) measurements. XPS results showed that the coating layer was composed of  $\text{LiBO}_2$ . The 2 wt% LBO-coated NMC exhibited the best electrochemical performance at high voltage (4.5 V). It was achieved a high capacity retention of 91.23% after 50 cycle which was 53.19% for the uncoated one. The rate capability of the coated electrode was improved obviously, exhibiting  $114.71 \text{ mAh}\cdot\text{g}^{-1}$  discharge capacity at 10 C. The interfacial resistance and charge transfer resistance was also decreased owing to the fact that the coating layer played a “firewall” between the electrolyte and the NMC electrode.

**Keywords:** cathode materials; surface coating; high cutoff voltage;  $\text{LiBO}_2$

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## 1. Introduction:

Recently, lithium-ion batteries (LIBs) developed fast and widely due to the applications in electric vehicles (EVs) and other electronic devices [1]. As we all knew, cathode material is the bottleneck of the LIBs' development [2]. Improving the capacity of cathode is the key point to meet the demand of the high performance LIBs. In order to reach the requirements of the state policy, especially for green energy EVs, we need to further improve their energy densities. Among the commercial cathode materials, layered oxide  $\text{LiNi}_x\text{Mn}_y\text{Co}_z\text{O}_2$  materials are one of the most promising materials because of their high capacities (especially for Ni-rich materials) and low cost [3, 4], which are far greater than  $\text{LiFePO}_4$  and spinel  $\text{LiMn}_2\text{O}_4$  [5, 6] in the prospect of practical applications. However, the current reversible capacities of Ni-rich (Ni content  $\geq 0.5$ ) materials are very limited if only charged to 4.3 V, which is lower than their theoretical capacity ( $278 \text{ mAh}\cdot\text{g}^{-1}$ ) [7]. It means that there is still a huge space for further improvement of the reversible capacity by improving the charge voltage.

Many researchers tried to obtain higher capacity by improving the operation voltage, unfortunately, obvious side reactions generally happened at the cathode electrode and electrolyte interface [8-10], which leads to form a thick solid-state electrolyte interphase (SEI) layer and consequently increases the interfacial resistance. It thus showed poor cycle performance, bad thermal stability at high voltage, which makes it urgent to find a solution to stabilize the interfacial structure at high voltage.

Surface coating is a useful method to solve the problem by stabilizing solid-liquid interface with inert metal-oxides [11-13]. So far, many metal-oxides were used as the

coating layers, such as  $\text{Al}_2\text{O}_3$ ,  $\text{TiO}_2$ ,  $\text{ZrO}_2$  [14],  $\text{MgO}$  [15],  $\text{ZnO}$  [16]. Lv et al. [17] used sol-gel method to build  $\text{Al}_2\text{O}_3$  layer on  $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$  surface, it showed good cycle performance; Chen et al. [18] reported that the improved cycle stability of  $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$  owing to the fact of surface treated with  $\text{TiO}_2$ . The enhancement of cycle performance and thermal stability is because that the metal oxide acts a protective layer preventing the materials from acidic liquid electrolyte [19]. However, the oxide layer also acts as an ion resistance because of the low electronic conductivity, which limited the rate capability and discharge capacity of the battery [7].

Li-ion conductor materials, such as  $\text{LiAlO}_2$ ,  $\text{Li}_2\text{TiO}_3$  [20],  $\text{Li}_2\text{ZrO}_3$  [21] were applied as the coating layers as well. For example, Zhang et al. [22] used hydrolysis-hydrothermal method to introduce  $\text{LiAlO}_2$  onto the surface of  $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$  (NMC), which played as a protective layer improving the cycle performance and enhancing the rate capability.  $\text{LiBO}_2$  is a kind of ideal coating material due to its chemical inert property in organic electrolyte. It is advantageous when compared with other high lithium content borate glasses because of the low viscosity which reduces hygroscopicity and carbonate retention [23]. We expect that it could make a uniform layer of  $\text{LiBO}_2$  on the surface of the layer cathode material to obtain good electrochemical performance at a high voltage of 4.5 V.

In our present work, we take a two-step process to coat  $\text{LiBO}_2$  layer on the surface of the layered oxide NMC material. After coating, the capacity retention is about 91.23% after 50 cycles, while that of bare sample is only 53.19%. When current density increases up to 10 C, it delivers a superior rate capability of  $114.71 \text{ mAh}\cdot\text{g}^{-1}$ , which is almost two times higher than that of bare sample. Also, it exhibits a good thermal stability in elevated

temperature with capacity retention of 80.7% after 100 cycles at 55 °C, which indicated that the coating material has good safety performance. The LBO layer plays the “firewall” to prevent the attack of acid from the electrolyte at high voltage especially at elevated temperature.

## 2. Experimental

### 2.1 Material synthesis

The NMC-layered materials were obtained by solid-state reaction method using the commercial transition-metal hydroxide precursors  $(\text{Ni}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2})(\text{OH})_2$  and  $\text{Li}_2\text{CO}_3$  as the pristine materials with weight ratio of 2:1.04. The mixture was heated at 450 °C for 5 h, and then calcined at 900 °C for 24 h in air.

The LBO-coated samples named LBO1 (1 wt% LBO), LBO2 (2 wt% LBO), and LBO3 (4 wt% LBO) were synthesized with the following process: firstly, a desired amount of  $\text{H}_3\text{BO}_3$  was dissolved in 50 mL ethanol, then 1g fresh NMC material was added into the solution. After supersonically dispersed for 30 min, a stoichiometric ratio ( $\text{B}:\text{Li} = 1:1$ )  $\text{LiOH}$  solution dropped slowly into the above ethanol solution. The mixture solution was stirred vigorously for 3 h to disperse evenly, and then was heated to 80 °C for 24 h to evaporate the solvent thoroughly. The obtained powders were finally calcined in 400 °C for 4 h with a  $3\text{ }^\circ\text{C}\cdot\text{min}^{-1}$  heating rate.

### 2.2 Material characterizations

Crystal structure of the samples was characterized with X-ray diffraction (XRD, Rigaku Ultima IV, Japan) using  $\text{Cu K}\alpha$  radiation. The scanning range was from  $2\theta = 10^\circ$  to  $80^\circ$  with the scan rate of  $10\text{ }^\circ\cdot\text{min}^{-1}$ .

Surface morphology of the materials was observed using scanning electron microscope (SEM 4800, Hitachi, Japan) with accelerating voltage of 20 kV and the transmission electron microscope (TEM, JEM 1400, JEOL, Japan). The composition and valence state distribution of the materials were identified by X-ray photoelectron spectroscopy (XPS, PHI 5000 Versa Probe III, ULVAC-PHI, Japan).

### 2.3 Electrochemical measurement

The working electrodes were prepared by slurry coating on the aluminum current collectors. The slurry consisted of 85wt% NMC cathode materials, 10 wt% conductive agent (super-P), and 5 wt% poly (vinylidene difluoride) binder in N-methylpyrrolidinone solvent. The as-prepared electrodes were dried in a vacuum oven at 120 °C for 16 h to remove the solvent and water and then pressed with 15 MPa. The amount of active material on each electrode is about 2.0 mg·cm<sup>-2</sup>. CR2016 coin-type cells were used for the electrochemical tests using lithium metal foil as the count electrodes, and LiPF<sub>6</sub> (1M) in a 1:1:1 (v:v:v) mixture of ethylene carbonate (EC), ethyl methyl carbonate (EMC), and dimethyl carbonate (DMC) as the electrolyte. The assemble procedure was carried out in an Ar-filled glove box with the contents of moisture and oxygen level less than 5 ppm. The galvanostatic charge-discharge tester (Neware, Shenzhen, China) was used for examining the electrochemical properties of the electrodes. The test voltage window was from 3.0 V to 4.5 V versus Li<sup>+</sup>/Li electrode at room temperature. The cyclic voltammetry was carried out between 3.0 V and 4.5 V on the electrochemical workstation (VMP2, Bio-Logic, France). The electrochemical impedance spectroscopy (EIS) was measured by an impedance analyzer (IM6, Zahner,

Germany) using the two-electrode cell mode. The frequency range is  $10^5$  Hz to  $10^{-2}$  Hz with a 5 mV amplitude voltage.

### 3. Results and discussion

X-ray diffraction patterns of the bare and LBO-coated NMC samples are shown in Fig.1. All the patterns are well fitted with a layered hexagonal  $\alpha$ -NaFeO<sub>2</sub> structure and belong to the space group  $R\bar{3}m$ . The (006)/(102) and (108)/(110) split peaks are clearly seen at about  $2\theta = 38.1^\circ$  and  $64.5^\circ$ , which indicates the high crystallization of NMC samples [24]. Comparing the XRD patterns for all the samples, there is no new phase formed due to the coating of LBO, which indicates that the coating process does not influence the crystal structure of NMC. The result also demonstrates that the coated LBO is in low amounts, as it does not show any XRD lines. The intensity ratio of (003) by (104) peak is larger than 1.2, which implies the low degree of ion disorders between  $\text{Li}^+$  and  $\text{Ni}^{2+}$ . The result points out that the coatings are just on the surface of the NMC materials rather than inserted into the cell structure.

In order to confirm the chemical composition of the surface coatings on the samples, XPS measurement has been carried out for the bare and LBO2 samples. The XPS analysis results are shown in Fig. 2. As shown in Fig.2a. The chemical condition of  $\text{O}^{2-}$  is analyzed. The O 1s peak at 529.30 eV is belong to the lattice oxygen of the host structure (M-O) [25], and the peak at 531.50 eV is attributed to the surface impurity ( $\text{Li}_2\text{CO}_3$ ) and/or surface adsorbed  $\text{OH}^-$  species. After coating, the peak shift to 531.15 eV and becoming stronger than the bare sample, which implies that the adsorbed oxygen species increase after surface coatings. At the same time, the existence of  $\text{B}^{3+}$  is identified on the LBO2

samples. As shown in Fig. 2b, the peak at 191.74 eV is consistent with the value reported for  $B^{3+}$  in  $LiBO_2$  [23].

After coating with LBO, the peak intensities of Ni, Co, and Mn elements are much weaker than those of the bare sample, which indicates that the LBO films have been successfully covered on the NMC surface, as XPS is a surface sensitive technique. As seen in Fig. 2c, the spectra of Ni  $2p_{3/2}$  indicate the co-existence of  $Ni^{2+}$  and  $Ni^{3+}$  on the surface, whose binding energies are about 854.15 eV and 855.55 eV, respectively, which well agree with the reported values for NMC [26]. The binding energy for Co  $2p_{3/2}$  is 779.75 eV, corresponding to  $Co^{3+}$  ions. As shown in Fig. 2e, the peak located at 642.15 eV confirms the presence of  $Mn^{4+}$  ions in the samples [27].

Surface morphology of the samples observed by SEM is shown in Fig. 3a-3d. The samples display well-crystallized particles, which consist of many primary particles. The size of primary particle is about 500 nm. We can observe difference intuitively between the bare and LBO-coated samples. The particle outlines of the LBO-coated samples are becoming unclear and rough with the growth of the coating amount, while the bare sample are angular and smooth. Obviously, the observation of LBO3 sample is almost the coated LBO layer (Fig. 3d), as the NMC particles are hardly seen. Comparing the TEM images of the bare and LBO2 samples shown in Fig. 3e and 3f, the fringe of the bare sample is smooth, however, the  $LiBO_2$  coating layer is clearly displayed as a compact film on the surface of NMC particles.

The initial charge/discharge curves of the samples between 3.0 V to 4.5 V at 25 °C and 0.2 C current density (1 C=160 mAh·g<sup>-1</sup>) are shown in Fig. 4a. The initial Coulombic



efficiencies are 82.52%, 83.37%, 84.40%, and 82.78%, respectively, for the bare, LBO1, LBO2, and LBO3 samples. The LBO coated samples show little higher Coulombic efficiency than the bare one, which indicates the suppression of parasitic reactions on the solid-liquid interface. The reason may be that the  $\text{LiBO}_2$  layer restrains the formation of  $\text{Li}_2\text{CO}_3$  or  $\text{LiOH}$  on the electrode surface and thus protect NMC from erosion by acids produced from electrolyte during cycling at high voltage[28]. The following cycle performance at 0.2 C for different samples and detailed comparison are shown in Fig. S1 and Table S1.

Fig. 4b shows the rate performance of the bare, LBO1, LBO2, and LBO3 electrodes ranging from 0.2 to 10 C. All of the discharge capacities decrease with the increasing current density. Among them, the LBO2 shows the best rate capability. The discharge capacities of the LBO2 material at 0.2, 0.5, 1, 2, 5, and 10 C are 184.38, 175.71, 167.01, 156.17, 137.65, and 114.71  $\text{mAh}\cdot\text{g}^{-1}$ . When the current density is shifted back to 0.2 C, its discharge capacity returns to 179.31  $\text{mAh}\cdot\text{g}^{-1}$ , which is 97.25% of the initial discharge capacity. However, the discharge capacities of the bare sample are 191.37, 163.47, 139.18, 120.37, 92.41, and 53.77  $\text{mAh}\cdot\text{g}^{-1}$  at the same conditions with the LBO2 sample. The capacity retention of the discharge capacity returns to 0.2 C is 82.28% of the initial discharge capacity. As displayed in Fig. 4c, if only comparing the rate performances of the bare and LBO2 samples, it is obvious that the LBO2 electrode delivers a superior rate capability, which is almost two times higher than that of the bare sample at high current densities. We can therefore infer that coating promotes the electrochemical performances of the electrodes.

The cycling performance of the samples is verified at 1 C current density. As shown in Fig. 4d, after 100 cycles, the LBO2 sample shows a discharge capacity of  $133.93 \text{ mAh}\cdot\text{g}^{-1}$  with capacity retention of 78.24%, while the capacity retention of the bare one is only 57.76% with a discharge capacity of  $93.11 \text{ mAh}\cdot\text{g}^{-1}$ . However, it should be pointed out that the electrochemical performance becomes worse, the presence of excess coating material could make the electronic conductivity lower and hinder the electrolyte wetting between particles, which will affect the lithium ion and electrons diffusion. Hence, when the coating amount of  $\text{LiBO}_2$  increases to 4 wt% (LBO3), its cycling performance is inferior to the LBO1 and LBO2 samples, even not good as the bare sample. The results suggest that the appropriate amount coating layer of  $\text{LiBO}_2$  can play the role of guard for the NMC grains to resist the corrosion by electrolyte even at high rate and at the same time allow the easy transport of lithium ions and electrons.

To gather information about the redox process during the charge and discharge processes. The cyclic voltammograms (CV) of the samples without and with  $\text{LiBO}_2$  coating have been logged with a scanning rate of  $0.1 \text{ mV}\cdot\text{s}^{-1}$  at  $25^\circ\text{C}$ . The comparison for the first four cycles is shown in Fig. 5a and 5b. Only one pair of redox peaks is observed in the CV profiles of the bare and LBO2 samples, which is related to the  $\text{Ni}^{2+}/\text{Ni}^{4+}$  redox process. There is no other peak observed corresponding to  $\text{Ni}^{2+}/\text{Ni}^{3+}$  conversion because of the Jahn-Teller distortion of  $\text{Ni}^{3+}$  in the octahedral structure [29, 30]. The sharp redox peaks of  $\text{Ni}^{2+}/\text{Ni}^{4+}$  can be observed at 3.8 V for both the bare and LBO2 samples. For the bare sample, the first oxidation peak is at 3.982 V, which is related to the delithiation process from the NMC lattice. The reduction peak is at 3.696 V, corresponding to the

lithiation into the lattice. The oxidation peak of the LBO2 sample is centered at 3.901 V, which is lower than that of bare sample. The reduction peak is centered at 3.698 V, which is higher than that of the bare sample. The difference of the redox peak potential  $\Delta V$  is 0.203 V, while the difference of the bare one is 0.286 V. the lower  $\Delta V$  means less polarization because the LBO layer suppresses the parasitic reactions between the electrolyte and NMC electrode, which implies that the coating layer improves the reversibility and reactivity of the redox reactions. After the first cycle, the oxidation of both electrodes shift to lower voltage and the peak current decreases, indicating the fading of capacity. While for the coated electrode the peak potential and intensity are steady after the 2<sup>nd</sup> cycle.

The electrochemical impedance spectroscopy (EIS) is used to clarify the difference of electrochemical polarization and ohmic polarization between the samples without and with coating during the cycling process. All of the testing cells were charged at 0.2 C via the constant current and constant potential procedures, and then discharged by a constant current procedure for 2 and 50 cycles. Then the cells were charged to 4.5 V with constant potential procedure for 30 min after constant current process for depolarization. The EIS tests were carried out in the range from  $10^5$  Hz to  $10^{-2}$  Hz with amplitude of 5 mV. As shown in Fig. 5c and 5d. Each of the Nyquist plots consist of two semicircles in the high-frequency and medium-frequency regions, and the equivalent circuit is shown in the inset of Fig. 4c.  $R_s$  is related to the impedance of solution. The first semicircle in high frequency is described as  $R_f$  which is relate to the surface film resistance (similar to SEI layer in anode) [31, 32]. The second semicircle is corresponding to the charge transfer

resistance of the solid-liquid interface [33, 34]. CPE (constant phase element) is described as the double layer capacity or passivation film capacitance.

The resistance values are listed in Table 1 after fitting. As we can see in the table, all the samples show low solution resistance of 2 to 4  $\Omega$ , which is almost negligible. After two cycles, the  $R_f$  values of the bare, LBO1, LBO2, and LBO3 samples are 91.12, 88.59, 87.62, and 95.02 $\Omega$ , which are close to each other. But, it is obvious that the  $R_{ct}$  values of coated samples are quite lower than that of the bare sample. The value of the LBO2 sample is smallest among of them. The  $R_{ct}$  value of LBO2 sample at the 2<sup>nd</sup> cycle is 93.81  $\Omega$ , while that of the bare sample is 192.10  $\Omega$ . After 50 cycles, the contrast of the  $R_{ct}$  values between bare and coated samples becomes larger. The  $R_{ct}$  value of bare sample increases to 803.41  $\Omega$ , leading to a poor rate capability, while that of the LBO sample is only 200.13  $\Omega$ , which is one fourth of the bare one. The  $R_{ct}$  value of LBO2 is also lower than that of other amount coatings. It should be pointed out that the  $R_s$  value of the LBO-coated sample decreases after 50 cycles, which is due to the instability of the surface film on the cathode materials [35, 36].

In previous reports, the  $Ni^{4+}$  ions can accelerate the oxidation of the electrolyte due to the high reactivity in electrolyte during the delithiation state of the  $Li_{1-x}Ni_{0.5}Mn_{0.3}Co_{0.2}O_2$  material. It also makes the phase transformation from the rhombohedral phase to spinel and rock salt phases by triggering the oxygen evolution from the surface of the material [37]. Based on the fact of EIS results companying with the electrochemical performance of the bare and LBO-coated samples, we could conclude that the LBO-layer inhibits the electrolyte decomposition at the solid-liquid interface and reduces the interface charge

transfer resistance[38].

In the Fig. 6, cycle performances of the bare and LBO2 cathodes at elevated temperature (55 °C) at 1 C are displayed. The initial discharge capacity of the bare and LBO2 are 180.31 and 188.53 mAh·g<sup>-1</sup>, which is higher than that at room temperature at the same rate due to the high temperature accelerating the diffusion speed of lithium ions. However, the bare sample shows poor electrochemical performance at high temperature, its capacity fades quickly. The discharge capacity of the bare sample is only 115.22 mAh·g<sup>-1</sup> after 100 cycles with a capacity retention of 63.9%. The capacity of LBO2 electrode decreases slowly and retains 80.7% of the initial capacity after 100 cycles. Based on the above results, we infer that the LBO coatings can server as protective layer to inhibit the side reactions between the electrolyte and NMC at elevated temperature and thus enhance the cycling performance of NMC materials and as well as the SEI layers.

#### 4. Conclusions:

In summary, LiNi<sub>0.5</sub>Mn<sub>0.3</sub>Co<sub>0.2</sub>O<sub>2</sub> was successfully coated by LiBO<sub>2</sub> with a simple method. LiBO<sub>2</sub> acts as the protect-layer to resist the side reactions at the solid-liquid interface; meanwhile, it enhances the structure stability and restrains the surface impedance during the cycling process. The cycle and rate tests showed that the LBO coatings enhanced the electrochemical performance of NMC cathode materials owing to the suppression of the parasitic reactions between the material and electrolyte. The 2 wt% LiBO<sub>2</sub>-coated (LBO2) LiNi<sub>0.5</sub>Mn<sub>0.3</sub>Co<sub>0.2</sub>O<sub>2</sub> exhibited much better rate capability and cyclability at elevated temperatures compared to the bare sample. The LBO2 sample presented significantly rate capability with a discharge capacity of 114.71 mAh·g<sup>-1</sup> when

current density is increased to 10 C. It also showed the enhancement of cycling performance at elevated temperature. The discharge capacity of the LBO2 sample is up to 152.17 mAh·g<sup>-1</sup> with a capacity retention of 80.7% after 100 cycles at 1C rate at 55 °C.

## Acknowledgments

We gratefully acknowledge the financial support from the National 973 Program (2015CB251102), the Key Project of NSFC (U1305246, 21673196, 21621091), the Fundamental Research Funds for the Central Universities (20720150042). Wu would like to thank the program for New Century Excellent Talents in University (NCET - 13 - 0879), the Natural Science Foundation of Fujian Province, China (Grant No. 2016J01069), and the “Min-Jiang Scholar” and “Tong-Jiang Scholar” programs.

## References

- [1] M. Armand, J.-M. Tarascon, Building better batteries, *Nature*, 451 (2008) 652-657.
- [2] F. Schipper, E.M. Erickson, C. Erk, J.-Y. Shin, F.F. Chesneau, D. Aurbach, Review—Recent Advances and Remaining Challenges for Lithium Ion Battery Cathodes I. Nickel-Rich, LiNi<sub>x</sub>Co<sub>y</sub>Mn<sub>z</sub>O<sub>2</sub>, *J. Electrochem. Soc.*, 164 (2017) A6220-A6228.
- [3] M.S. Whittingham, Lithium batteries and cathode materials, *Chemical reviews*, 104 (2004) 4271-4302.
- [4] B. Dunn, H. Kamath, J.-M. Tarascon, Electrical energy storage for the grid: a battery of choices, *Science*, 334 (2011) 928-935.
- [5] J.B. Goodenough, K.-S. Park, The Li-ion rechargeable battery: a perspective, *J. Am. Chem. Soc.*, 135 (2013) 1167-1176.
- [6] T. Ohzuku, R.J. Brodd, An overview of positive-electrode materials for advanced lithium-ion batteries, *J. Power Sources*, 174 (2007) 449-456.
- [7] W. Liu, P. Oh, X. Liu, M.J. Lee, W. Cho, S. Chae, Y. Kim, J. Cho, Nickel-Rich Layered Lithium Transition-Metal Oxide for High-Energy Lithium-Ion Batteries, *Angew. Chem. Inter. Ed.*, 54 (2015) 4440-4457.
- [8] B.C. Park, H.B. Kim, S.T. Myung, K. Amine, I. Belharouak, S.M. Lee, Y.K. Sun, Improvement of structural and electrochemical properties of AlF<sub>3</sub>-coated Li[Ni<sub>1/3</sub>Co<sub>1/3</sub>Mn<sub>1/3</sub>]O<sub>2</sub> cathode materials on high voltage region, *J. Power Sources*, 178 (2008) 826-831.
- [9] S.H. Ju, I.-S. Kang, Y.-S. Lee, W.-K. Shin, S. Kim, K. Shin, D.-W. Kim, Improvement of the cycling performance of LiNi<sub>0.6</sub>Co<sub>0.2</sub>Mn<sub>0.2</sub>O<sub>2</sub> cathode active materials by a dual-conductive polymer coating, *ACS Appl. Mater. Interfaces*, 6 (2014) 2546-2552.
- [10] X.-H. Liu, L.-Q. Kou, T. Shi, K. Liu, L. Chen, Excellent high rate capability and high voltage cycling stability of Y<sub>2</sub>O<sub>3</sub>-coated LiNi<sub>0.5</sub>Co<sub>0.2</sub>Mn<sub>0.3</sub>O<sub>2</sub>, *J. Power Sources*, 267 (2014) 874-880.

- [11] H. Konishi, M. Yoshikawa, T. Hirano, The effect of thermal stability for high-Ni-content layer-structured cathode materials,  $\text{LiNi}_{0.8}\text{Mn}_{0.1-x}\text{Co}_{0.1}\text{Mo}_x\text{O}_2$  ( $x = 0, 0.02, 0.04$ ), *J. Power Sources*, 244 (2013) 23-28.
- [12] Y.-S. Lee, W.-K. Shin, A.G. Kannan, S.M. Koo, D.-W. Kim, Improvement of the Cycling Performance and Thermal Stability of Lithium-Ion Cells by Double-Layer Coating of Cathode Materials with  $\text{Al}_2\text{O}_3$  Nanoparticles and Conductive Polymer, *ACS Appl. Mater. Interfaces*, 7 (2015) 13944-13951.
- [13] C. Li, H.P. Zhang, L.J. Fu, H. Liu, Y.P. Wu, E. Rahm, R. Holze, H.Q. Wu, Cathode materials modified by surface coating for lithium ion batteries, *Electrochimica Acta*, 51 (2006) 3872-3883.
- [14] J.-Z. Kong, S.-S. Wang, G.-A. Tai, L. Zhu, L.-G. Wang, H.-F. Zhai, D. Wu, A.-D. Li, H. Li, Enhanced electrochemical performance of  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  cathode material by ultrathin  $\text{ZrO}_2$  coating, *J. Alloys Comp.*, 657 (2016) 593-600.
- [15] J.-H. Shim, S. Lee, S.S. Park, Effects of  $\text{MgO}$  coating on the structural and electrochemical characteristics of  $\text{LiCoO}_2$  as cathode materials for lithium ion battery, *Chem. Mater.*, 26 (2014) 2537-2543.
- [16] J.-Z. Kong, C. Ren, G.-A. Tai, X. Zhang, A.-D. Li, D. Wu, H. Li, F. Zhou, Ultrathin  $\text{ZnO}$  coating for improved electrochemical performance of  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  cathode material, *J. Power Sources*, 266 (2014) 433-439.
- [17] Q. Qiu, X. Huang, Y. Chen, Y. Tan, W. Lv,  $\text{Al}_2\text{O}_3$  coated  $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$  cathode material by sol-gel method: Preparation and characterization, *Ceramics International*, 40 (2014) 10511-10516.
- [18] Y. Chen, Y. Zhang, B. Chen, Z. Wang, C. Lu, An approach to application for  $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$  cathode material at high cutoff voltage by  $\text{TiO}_2$  coating, *J. Power Sources*, 256 (2014) 20-27.
- [19] W. Liu, M. Wang, X.L. Gao, W. Zhang, J. Chen, H. Zhou, X. Zhang, Improvement of the high-temperature, high-voltage cycling performance of  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  cathode with  $\text{TiO}_2$  coating, *J. Alloys Compounds*, 543 (2012) 181-188.
- [20] J. Wang, Y. Yu, B. Li, T. Fu, D. Xie, J. Cai, J. Zhao, Improving the electrochemical properties of  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  at 4.6 V cutoff potential by surface coating with  $\text{Li}_2\text{TiO}_3$  for lithium-ion batteries, *Phys. Chem. Chem. Phys.*, 17 (2015) 32033-32043.
- [21] D. Wang, X. Li, W. Wang, Z. Wang, H. Guo, J. Ru, Improvement of high voltage electrochemical performance of  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  cathode materials via  $\text{Li}_2\text{ZrO}_3$  coating, *Ceramics International*, 41 (2015) 6663-6667.
- [22] Z. Chen, Q. Zhang, M. Xu, X. Zhou, H. Zhu, K. Zhang, A hydrolysis-hydrothermal route for the synthesis of ultrathin  $\text{LiAlO}_2$ -inlaid  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  as a high-performance cathode material for lithium ion batteries, *J. Mater. Chem. A*, 3 (2015) 894-904.
- [23] B. Chowdari, Z. Rong, The role of  $\text{Bi}_2\text{O}_3$  as a network modifier and a network former in  $x\text{Bi}_2\text{O}_3 \cdot (1-x)\text{LiBO}_2$  glass system, *Solid State Ionics*, 90 (1996) 151-160.
- [24] K.M. Shaju, G.V. Subba Rao, B.V.R. Chowdari, Performance of layered  $\text{Li}(\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3})\text{O}_2$  as cathode for Li-ion batteries, *Electrochim. Acta*, 48 (2002) 145-151.
- [25] J.-Z. Kong, H.-F. Zhai, C. Ren, G.-A. Tai, X.-Y. Yang, F. Zhou, H. Li, J.-X. Li, Z. Tang, High-capacity  $\text{Li}(\text{Ni}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3})\text{O}_2$  lithium-ion battery cathode synthesized using a green chelating agent, *J. Solid State Electrochem.*, 18 (2014) 181-188.
- [26] Y.-M. Lee, K.-M. Nam, E.-H. Hwang, Y.-G. Kwon, D.-H. Kang, S.-S. Kim, S.-W. Song, Interfacial origin of performance improvement and fade for 4.6 V  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  battery cathodes, *J. Phys. Chem. C*, 118 (2014) 10631-10639.
- [27] X. Yang, R. Yu, L. Ge, D. Wang, Q. Zhao, X. Wang, Y. Bai, H. Yuan, H. Shu, Facile synthesis and performances of nanosized  $\text{Li}_2\text{TiO}_3$ -based shell encapsulated  $\text{LiMn}_{1/3}\text{Ni}_{1/3}\text{Co}_{1/3}\text{O}_2$  microspheres, *J. Mater.*

Chem. A, 2 (2014) 8362-8368.

- [28] G. Hu, M. Zhang, L. Wu, Z. Peng, K. Du, Y. Cao, Effects of  $\text{Li}_2\text{SiO}_3$  coating on the performance of  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  cathode material for lithium ion batteries, *J. Alloys Compounds*, 690 (2017) 589-597.
- [29] S.H. Park, C.S. Yoon, S.G. Kang, H.S. Kim, S.I. Moon, Y.K. Sun, Synthesis and structural characterization of layered  $\text{Li}[\text{Ni}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}]\text{O}_2$  cathode materials by ultrasonic spray pyrolysis method, *Electrochim. Acta*, 49 (2004) 557-563.
- [30] J. Ni, H. Zhou, J. Chen, X. Zhang, Improved electrochemical performance of layered  $\text{LiNi}_{0.4}\text{Co}_{0.2}\text{Mn}_{0.4}\text{O}_2$  via  $\text{Li}_2\text{ZrO}_3$  coating, *Electrochim. Acta*, 53 (2008) 3075-3083.
- [31] J.B. Goodenough, Y. Kim, Challenges for rechargeable Li batteries, *Chem. Mater.*, 22 (2009) 587-603.
- [32] P. Li, H. Liu, B. Lu, Y. Wei, Formation mechanism of 1D ZnO nanowhiskers in aqueous solution, *J. Phys. Chem. C*, 114 (2010) 21132-21137.
- [33] G. Yan, X. Li, Z. Wang, H. Guo, C. Wang, Tris(trimethylsilyl)phosphate: A film-forming additive for high voltage cathode material in lithium-ion batteries, *J. Power Sources*, 248 (2014) 1306-1311.
- [34] J. Liu, A. Manthiram, Functional surface modifications of a high capacity layered  $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$  cathode, *J. Mater. Chem.*, 20 (2010) 3961-3967.
- [35] S. Shi, J. Tu, Y. Tang, X. Liu, Y. Zhang, X. Wang, C. Gu, Enhanced cycling stability of  $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$  by surface modification of MgO with melting impregnation method, *Electrochim. Acta*, 88 (2013) 671-679.
- [36] L.J. Xi, H.-E. Wang, Z.G. Lu, S.L. Yang, R.G. Ma, J.Q. Deng, C.Y. Chung, Facile synthesis of porous  $\text{LiMn}_2\text{O}_4$  spheres as positive electrode for high-power lithium ion batteries, *J. Power Sources*, 198 (2012) 251-257.
- [37] S.K. Jung, H. Gwon, J. Hong, K.Y. Park, D.H. Seo, H. Kim, J. Hyun, W. Yang, K. Kang, Understanding the degradation mechanisms of  $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$  cathode material in lithium ion batteries, *Adv. Energy Mater.*, 4 (2014) 94-98.
- [38] K. Edström, T. Gustafsson, J.O. Thomas, The cathode–electrolyte interface in the Li-ion battery, *Electrochim. Acta*, 50 (2004) 397-403.



**Table and Figures Captions:**

Table 1 Impedance data of the bare and LBO2 samples after different number of cycles.

Fig. 1 XRD patterns of as prepared four samples.

Fig. 2 XPS spectra of a) O 1s, b) B 1s, c) Ni 2p, d) Co 2p, and e) Mn 2p for the bare and LBO2 samples.

Fig. 3 SEM images of the a) bare, b) LBO1, c) LBO2, and d) LBO3 samples; TEM image of the e) bare and f) LBO2 samples.

Fig. 4 Electrochemical performances of the different samples: a) initial cycle curves, b) rate capability, c) rate comparison of the bare and LBO2, d) cycle stability.

Fig. 5 The CV curves of the a) bare and b) LBO2 samples with scanning rate of 0.1 mV/s. EIS plot of the bare and LBO2 samples after c) 2, d) 50 cycles at 4.5 V.

Fig. 6 Cycle performance of the bare and LBO2 samples at 1C and 55°C.

Table 1 Impedance data of the bare and LBO2 samples after different number of cycles.

| Cycle number   | 2              |                |                 | 50             |                |                 |
|----------------|----------------|----------------|-----------------|----------------|----------------|-----------------|
| R ( $\Omega$ ) | R <sub>s</sub> | R <sub>f</sub> | R <sub>ct</sub> | R <sub>s</sub> | R <sub>f</sub> | R <sub>ct</sub> |
| bare           | 2.08           | 91.12          | 192.10          | 2.16           | 160.6          | 803.41          |
| LBO1           | 2.23           | 88.59          | 97.83           | 2.24           | 19.94          | 324.21          |
| LBO2           | 2.94           | 87.62          | 93.81           | 1.67           | 24.57          | 200.13          |
| LBO3           | 3.11           | 95.02          | 106.20          | 2.56           | 45.29          | 969.64          |

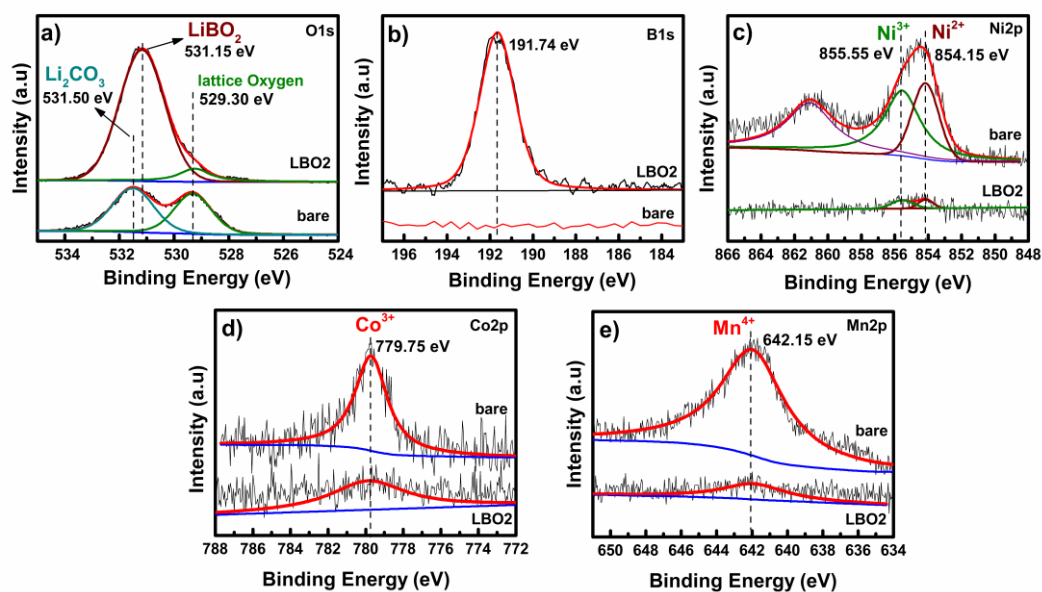


Fig. 2 XPS spectra of a) O 1s, b) B 1s, c) Ni 2p, d) Co 2p, and e) Mn 2p for the bare and LBO2 samples.

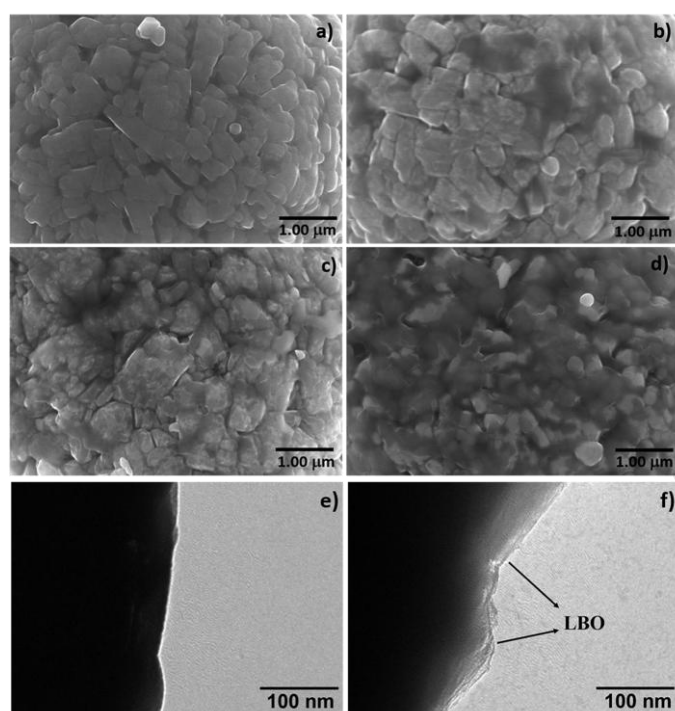


Fig. 3 SEM images of the a) bare, b) LBO1, c) LBO2, and d) LBO3 samples; TEM image of the e) bare and f) LBO2 samples.

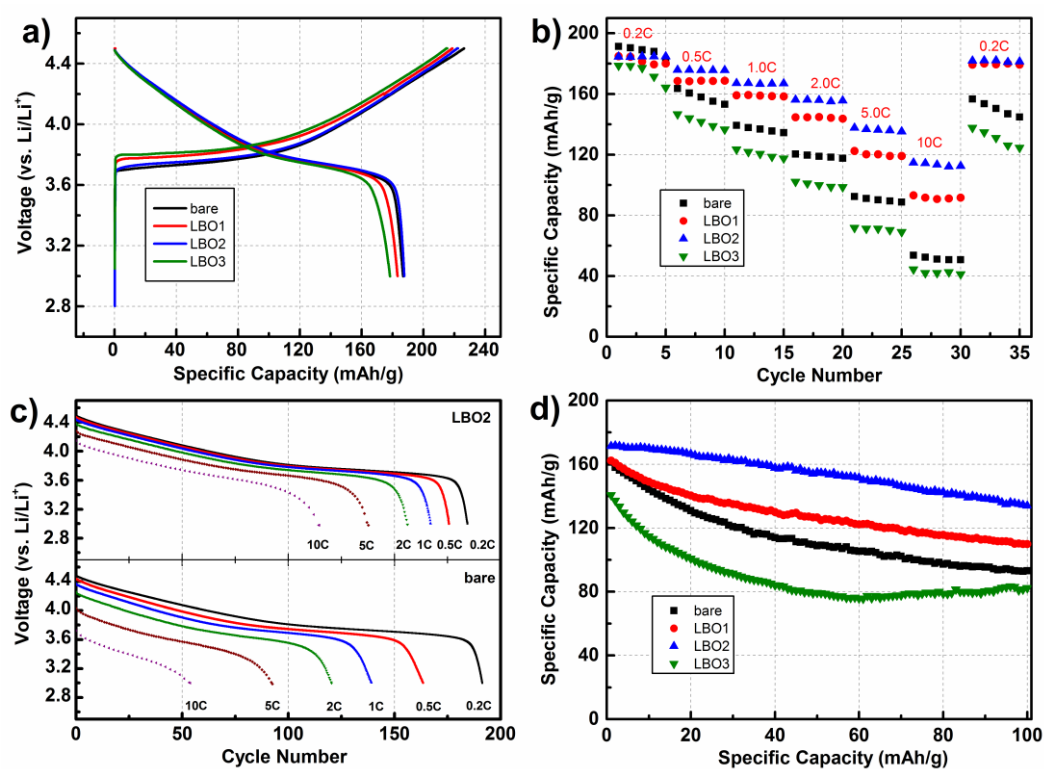


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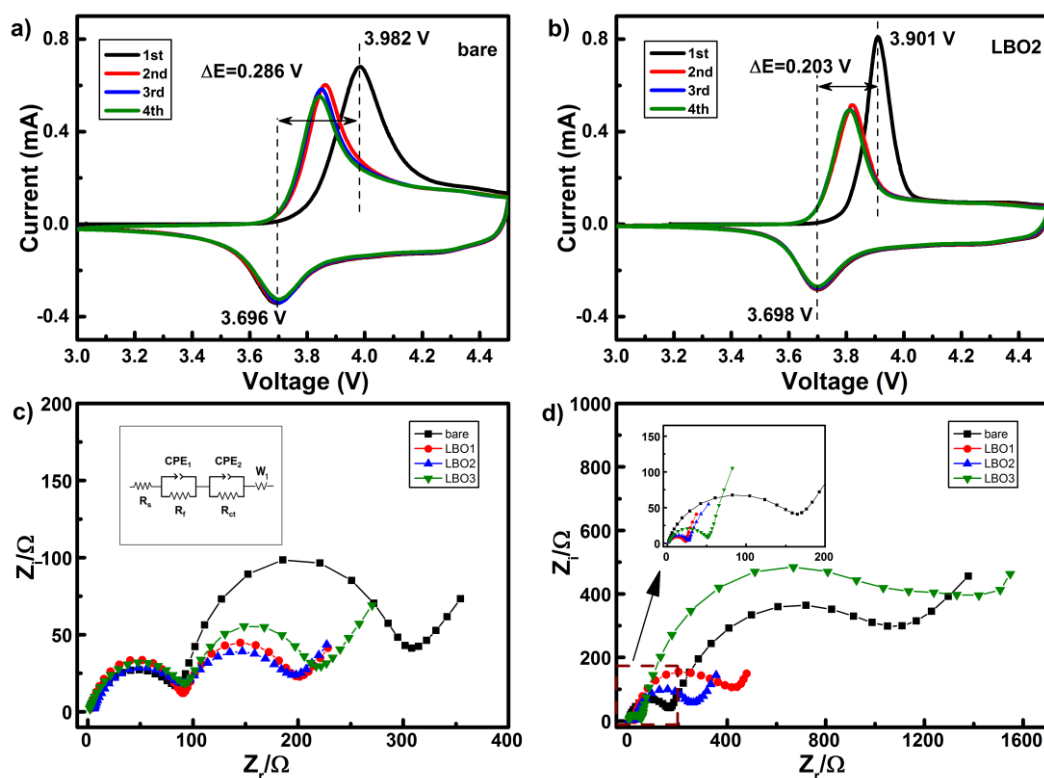


Fig. 5 The CV curves of the a) bare and b) LBO2 samples with scanning rate of 0.1 mV/s. EIS plot of the bare and LBO2 samples after c) 2, d) 50 cycles at 4.5 V.

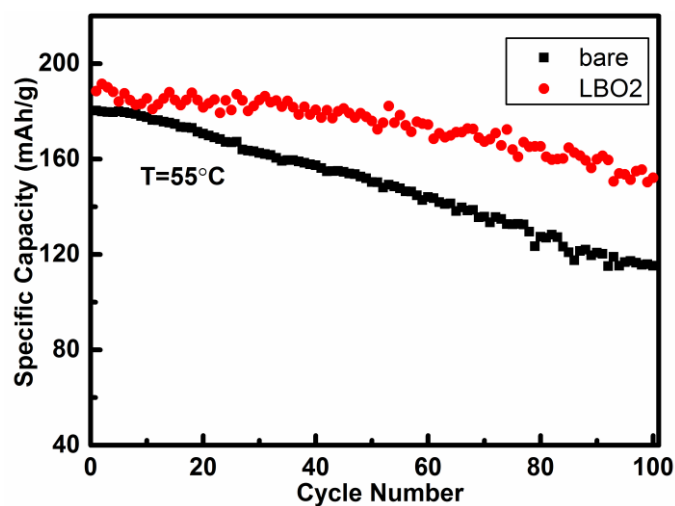


Fig. 6 cycle performance of the bare and LBO2 samples at 1C and 55°C.